

# Blockchain Based IoT Security: A Comprehensive Review of Architectures Challenges and Future Directions

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**Abstract**—Healthcare, transportation, and smart cities marked revolutionary growth in the market. They are increasingly relying on the Internet of Things (IoT) that may be appropriately identified with strong privacy and security threats. The decentralized, scalable function of IoT devices leaves them uniquely at risk of cyberattacks or breaches. Many security schemes depend on centralized approaches that just don't fit in this scenario. A technology like blockchain addresses existing security concerns by facilitating automation using smart contracts and secure communication. This article offers a coherent and thoughtful exposition on the growing security threats to IoT. It addresses gaps in IoT architecture and configuration within pseudo-omnichannels. The discussion further emphasizes latency, energy efficiency of connected devices, and the costs tied to rigid cyber infrastructures that may fall short. The impacts of security are examined using frameworks dependent on artificial intelligence, lightweight blockchains, and post-quantum cryptography as a basis of secure omnichannel architecture. Additionally, the framework addresses universal receipt control and gate access mechanisms as include parts of IoT security.

**Index Terms**—Blockchain; Internet of Things (IoT); Security; Smart Contracts; Decentralized Identity (DID)

## I. INTRODUCTION

### A. The Expanding IoT and Its Security Challenges

The Internet of Things (IoT) revolutionize areas such as healthcare, manufacturing, smart cities, transportation and agriculture. There were already nearly 18.8 billion IoT devices in use by 2020, with numbers expected to surpass 40 billion by the year 2030 [1]. This kind of boom speeds up innovation, but increases the difficulty ensuring security for IoT infrastructures. Traditional security models that rely on centralized hubs do not work well in such environments where IoT devices may be resource-constrained and have limited access to infrastructure.

### B. Motivation for Using Blockchain based IoT Security

Blockchain's distributed and immutable architecture provides perfect answers to the ongoing security dilemmas of IoT. Blockchain is a technology that offers tamper proof transaction capabilities as well as distributed consensus and auditable trust.

Basic functions such as cryptographic hashing, consensus algorithms and smart contracts enable secure peer-to-peer communication, traceability and non-tampering. Decentralized identity (DID) also better supports device authentication and trust, especially useful in supply chains, smart grids, or connected healthcare systems [2].

### C. Objectives and Scope

We systematically review the state-of-the-art of blockchain enabled IoT and Industrial IoT (IIoT) security frameworks. It also outlines significant threats including botnet attacks, side-channel leaks and firmware exploits and explores decentralized mitigation approaches.

New discoveries include AI hybrids raising threat detection rates by 40%, edge-IPFS cutting health data latency in half [3], and DIDLT-BC reducing authentication delays by 30% [4].

### D. Contributions

Our survey captures four main contribution themes to the blockchain-IoT security research:

- 1) **Legacy System Integration:** The majority of legacy IoT devices lack cryptographic verification processes, leaving modern smart factory implementations vulnerable. We looked at compatibility approaches and integration frameworks proposed in the literature and clarify how they aid in reducing security threats.